

Generative AI with Gemini

Total Duration: 6 Hours

Course Overview

This program is designed to introduce participants to the **Gemini family of models** and **Google AI Studio** for building practical AI-driven solutions. Through guided sessions and hands-on prototyping, learners will explore how to design, deploy, and optimize applications powered by state-of-the-art generative AI.

The course moves from **core concepts** to **applied development**, helping participants gain both **technical know-how** and the ability to create **real-world AI experiences**—ranging from chat assistants to multimodal apps.

Learning Outcomes

By the end of this program, participants will be able to:

- Explain the capabilities and unique features of **Google's Gemini models**.
 - Apply **prompt engineering strategies** to control outputs effectively.
 - Develop **chatbots, Q&A tools, and multimodal applications** using Gemini.
 - Integrate **retrieval-augmented generation (RAG)** for knowledge-grounded responses.
 - Prototype and publish **shareable AI applications** with frameworks like Streamlit and Gradio.
 - Translate AI concepts into solutions for industries such as education, business, and healthcare.
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Prerequisites

- Basic familiarity with **AI and machine learning concepts**.



- Comfort with **Python programming** is recommended.
 - Enthusiasm to experiment with **AI-driven creativity and applications**.
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Course Modules

Day 1 – Exploring Gemini and AI Studio

1.5 Hours

- What is Generative AI? How LLMs shape the AI ecosystem
 - Gemini models in focus: **Gemini 1.5 Pro, Gemini Flash, Gemini Nano**
 - Gemini's multimodal strengths: text, code, and images
 - Getting started in **Google AI Studio**:
 - Navigating the interface
 - Model selection and usage limits
 - Hands-on: Running your **first prompt** in AI Studio
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Day 1 – Creating Interactive Apps with Streamlit

1.5 Hours

- Introduction to Streamlit for AI-driven UIs
- Building a **simple chat interface** with Gemini via Python SDK
- Adding interactivity: sliders, controls, and session history
- Deploying your app: local runs & Streamlit Cloud
- Mini-project: **Q&A bot for course materials**

Day 2 – Rapid Prototyping with Gradio

1.5 Hours

- Gradio basics: blocks, components, and layouts
 - Designing a conversational app with Gemini
 - Building a multimodal prototype: **image input** → **text response**
 - Comparing **Streamlit vs Gradio**: when to use each
 - Publishing lightweight demos on **Hugging Face Spaces**
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Day 2 – Advanced Patterns: RAG & Multimodality

1.5 Hours

- Why RAG? Grounding responses with external knowledge
 - Using **Gemini embeddings** and connecting with **vector databases** (ChromaDB, FAISS)
 - Integrating with **LangChain** for retrieval workflows
 - Exploring **multimodal applications**: combining text + images
 - Deployment pathways: Google Cloud, AI Studio projects, or public demo platforms
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Tools & Setup

- **Software**: Python 3.x, Jupyter/VS Code, Google AI Studio, Streamlit/Gradio
 - **Hardware**: Laptop/Desktop (Core i5 or above, 8GB RAM minimum)
 - **Internet**: Stable broadband (30 Mbps+)
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SOFTWARE REQUIREMENT

- Python 3.x
- IDEs: PyCharm, Jupyter Notebook, or VS Code

HARDWARE REQUIREMENT

- Desktop / Laptop with Core i5 or higher Processor
- Windows 10 OS or Equivalent
- 8 GB of RAM
- Minimum 30 MBPS internet connection.